SUSPICIOUS FILEACCESS



CYBER FORENSICS CASE STUDY

KEY DEFINITIONS

Cyber Forensics: The scientific process of collecting, preserving, and analyzing digital evidence from computer systems and devices.

Suspicious File Access

Unauthorized or unusual interactions with files that indicate potential system compromise, data breach, or malicious activity.



Digital Evidence

Information stored or transmitted in digital form for legal proceedings.



Chain of Custody

Documentation tracking evidence handling to ensure integrity.



Legal Importance

Forensic processes ensure evidence is admissible in court.

INVESTIGATION SCOPE

Defining the boundaries and coverage of this forensic investigation to ensure a focused, systematic examination of the suspicious file access incident.



Single Device

Focus on single laptop owned by friend as primary evidence source.



System Logs

Examination limited to file system and system log analysis.



No Network Analysis

Network traffic and external devices excluded from this examination.



Timeframe

Incident window plus prior 30 days of activity analyzed.

INVESTIGATION OBJECTIVE

Defining the goals of our forensic investigation to ensure a thorough, methodical approach that yields actionable and legally admissible results.

Identify Access

Determine the extent and nature of suspicious file access events.

Determine Source

Identify if unauthorized user or malware was involved in the incident.

Preserve Evidence

Maintain evidence integrity and chain of custody for legal admissibility.

Clear Report

Provide comprehensive report for awareness and potential legal use.

IDENTIFICATION
SECURE THE SCENE

Key steps to secure the crime scene and gather volatile data before it's lost.



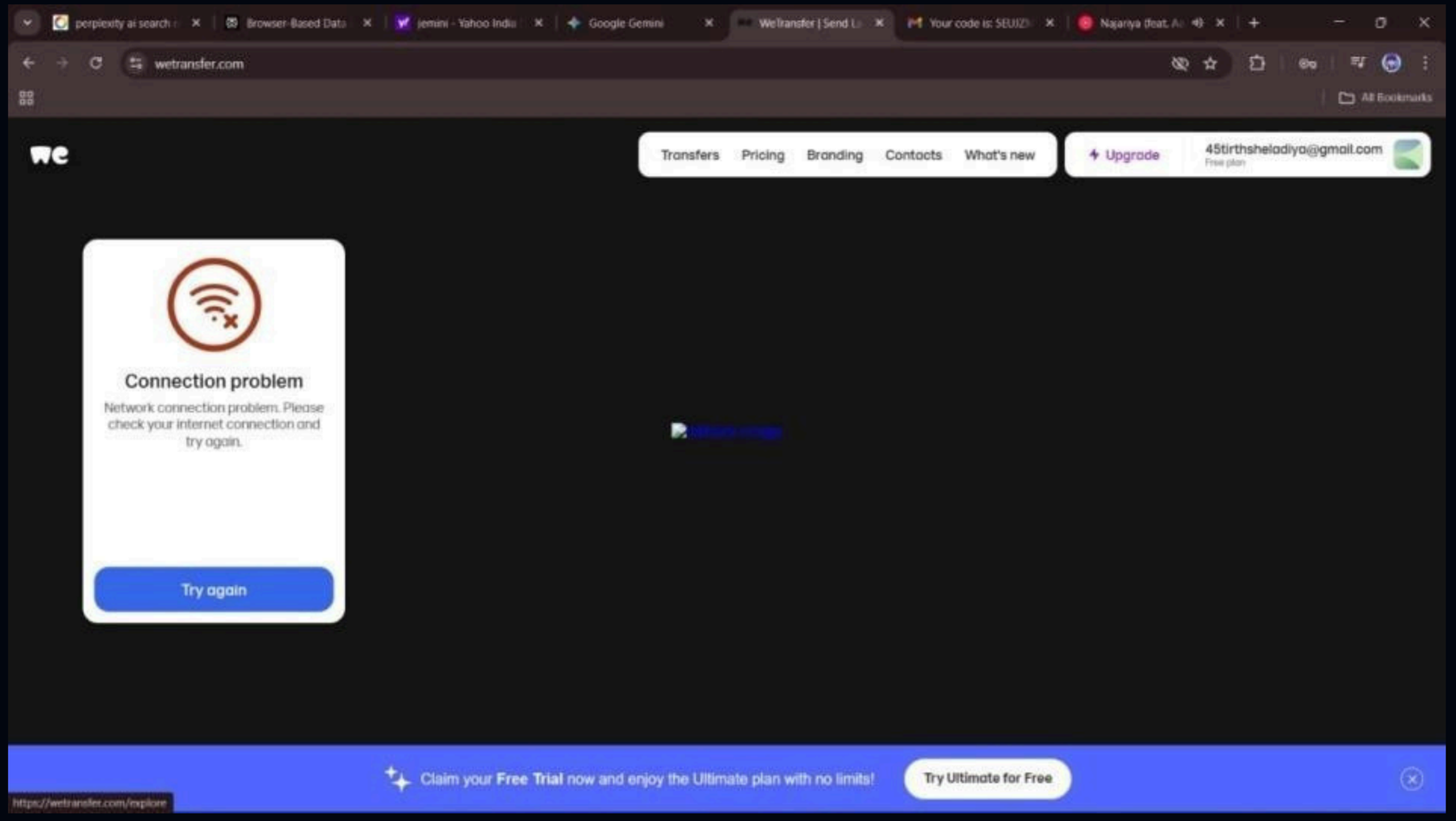
Crime Securing

Device Listing

Volatile Observation

Crime Securing

- System secured immediately after detection
- Network connectivity disabled
- Prevented further file modification
- Treated system as digital crime scene
- Preserved live evidence



Device Listing

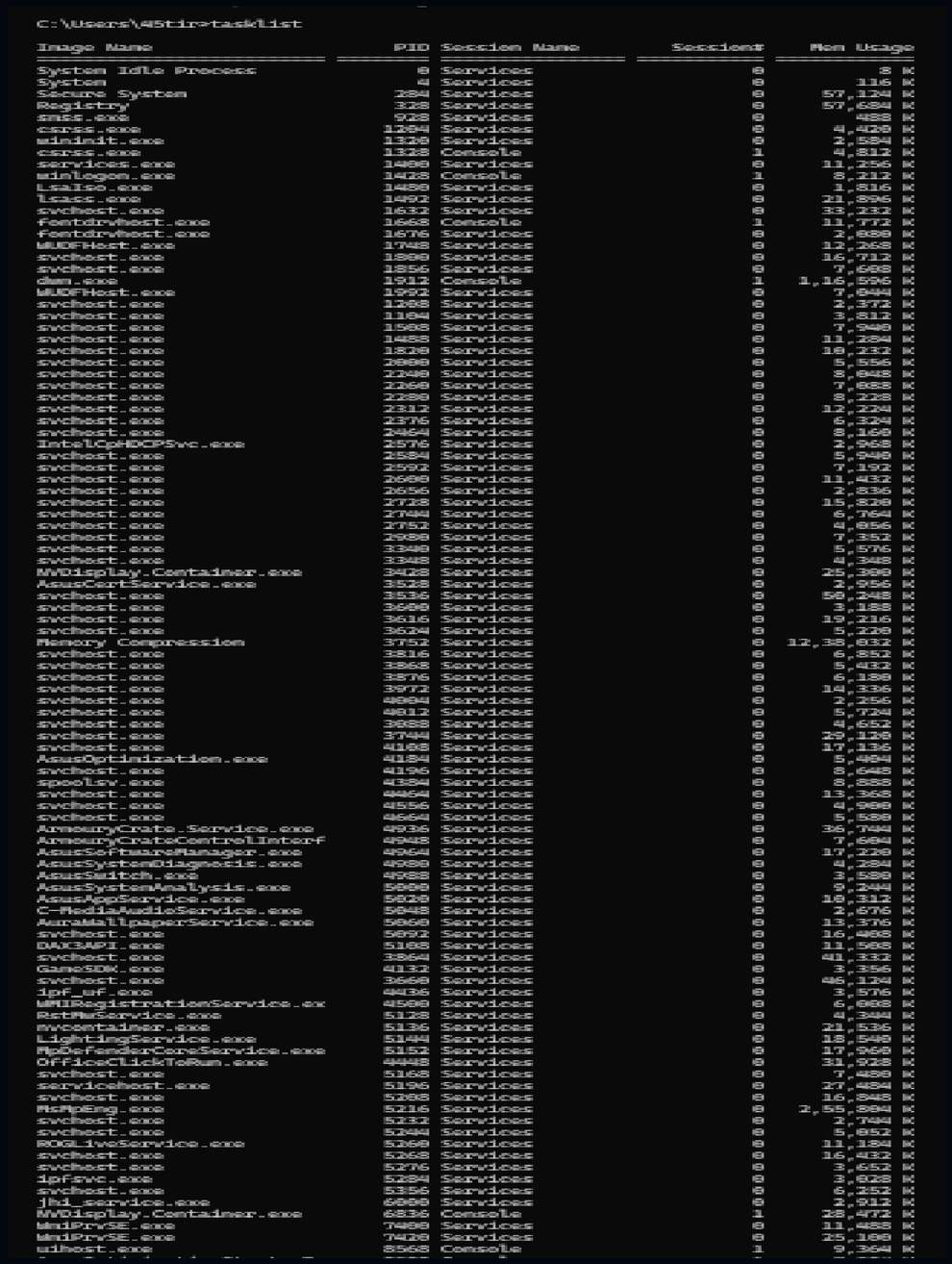
- Connected devices identified
- USB storage devices detected
- External devices analyzed
- Possible data transfer sources checked
- Samsung mobile device observed

```
PS C:\WINDOWS\system32> Get-PnpDevice | Where-Object { $_.FriendlyName -like "*USB*" } |
>> Select-Object FriendlyName, Class, InstanceId

FriendlyName                                                Class    InstanceId
------------                                                -----    ----------
SAMSUNG Mobile USB Remote NDIS Network Device #2            Net      USB\VID_04E8&PID_6863&RNDIS\6&1A95662B&1&0000
USB2.0 HD UVC WebCam                                        Camera   USB\VID_322E&PID_202C&MI_00\6&19B42EEC&0&0000
SAMSUNG Mobile USB Composite Device                         USB      USB\VID_04E8&PID_6863\R5CT72MAWKE
Unknown USB Device (Device Descriptor Request Failed)       USB      USB\VID_0000&PID_0002\5&242A2F40&0&1
Unknown USB Device (Device Descriptor Request Failed)       USB      USB\VID_0000&PID_0002\5&242A2F40&0&4
USB-Enhanced-SERIAL CH9102 (COM5)                           Ports    USB\VID_1A86&PID_55D4\5718007863
USB4 Virtual power coordination device                               USB4\VIRTUAL_POWER_PDO\4&8469D8&0&0
USB4 Root Router (1.0)                                      USB      USB4\ROOT_DEVICE_ROUTER&VID_8086&PID_463E\4...
USB Input Device                                            HIDClass USB\VID_046D&PID_C542\5&242A2F40&0&1
USB Input Device                                            HIDClass USB\VID_046D&PID_C542\5&242A2F40&0&2
Intel(R) USB 3.10 eXtensible Host Controller - 1.20 (Microsoft) USB  PCI\VEN_8086&DEV_51ED&SUBSYS_201F1043&REV_0...
USB Composite Device                                        USB      USB\VID_04D9&PID_A09F\5&242A2F40&0&1
SAMSUNG Mobile USB Modem                                    Modem    USB\VID_04E8&PID_6860&MODEM\6&1C5A7CE6&0&0001
SAMSUNG Mobile USB Composite Device                         USB      USB\VID_04E8&PID_6860\RZCT815F5ET
Intel(R) USB 3.20 eXtensible Host Controller - 1.20 (Microsoft) USB  PCI\VEN_8086&DEV_461E&SUBSYS_00000000&REV_0...
USB4(TM) Host Router (Microsoft)                            USB      PCI\VEN_8086&DEV_463E&SUBSYS_11112222&REV_0...
USB Composite Device                                        USB      USB\VID_322E&PID_202C\5&242A2F40&0&7
USB Root Hub (USB 3.0)                                      USB      USB\ROOT_HUB30\4&3AF0ECE5&0&0
USB Composite Device                                        USB      USB\VID_0B05&PID_6203\5&242A2F40&0&9
SAMSUNG Mobile USB Composite Device                         USB      USB\VID_04E8&PID_6860\R5CT72MAWKE
USB Input Device                                            HIDClass USB\VID_0B05&PID_6203&MI_03\6&E200585&0&0003
USB Input Device                                            HIDClass USB\VID_04D9&PID_A09F&MI_02\6&E7F607C&0&0002
USB Input Device                                            HIDClass USB\VID_04D9&PID_A09F&MI_01\6&E7F607C&0&0001
USB Root Hub (USB 3.0)                                      USB      USB\ROOT_HUB30\4&11BF4672&0&0
USB Input Device                                            HIDClass USB\VID_04D9&PID_A09F&MI_00\6&E7F607C&0&0000
```

Volatile Observation

- Live system activity observed
- Running applications identified
- Active browser sessions checked
- Logged-in user detected
- Network activity monitored



```
C:\Users\45tir>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0        116 K
Secure System                  184 Services                   0     57,124 K
Registry                       328 Services                   0     57,684 K
smss.exe                       928 Services                   0        488 K
csrss.exe                     1284 Services                   0      4,408 K
wininit.exe                   1320 Services                   0      2,584 K
csrss.exe                     1328 Console                    1      4,812 K
services.exe                  1400 Services                   0     11,256 K
winlogon.exe                  1428 Console                    1      8,212 K
LsaIso.exe                    1480 Services                   0      1,816 K
lsass.exe                     1492 Services                   0     21,896 K
svchost.exe                   1632 Services                   0     33,232 K
fontdrvhost.exe               1668 Console                    1     11,772 K
fontdrvhost.exe               1676 Services                   0      2,680 K
WUDFHost.exe                  1748 Services                   0     12,268 K
svchost.exe                   1808 Services                   0     16,712 K
svchost.exe                   1856 Services                   0      7,688 K
dwm.exe                       1912 Console                    1  1,16,596 K
WUDFHost.exe                  1952 Services                   0      7,044 K
svchost.exe                   1268 Services                   0      2,372 K
svchost.exe                   1184 Services                   0      3,812 K
svchost.exe                   1508 Services                   0      7,948 K
svchost.exe                   1488 Services                   0     11,284 K
svchost.exe                   1828 Services                   0     10,232 K
svchost.exe                   2008 Services                   0      5,856 K
svchost.exe                   2240 Services                   0      3,048 K
svchost.exe                   2260 Services                   0      7,088 K
svchost.exe                   2280 Services                   0      8,228 K
svchost.exe                   2312 Services                   0     12,224 K
svchost.exe                   2376 Services                   0      6,324 K
svchost.exe                   2464 Services                   0      8,160 K
IntelCpHDCPSvc.exe            2576 Services                   0      2,968 K
svchost.exe                   2584 Services                   0      5,940 K
svchost.exe                   2592 Services                   0      7,192 K
svchost.exe                   2600 Services                   0     11,432 K
svchost.exe                   2656 Services                   0      2,836 K
svchost.exe                   2728 Services                   0     15,820 K
svchost.exe                   2744 Services                   0      6,764 K
svchost.exe                   2752 Services                   0      4,856 K
svchost.exe                   2588 Services                   0      7,352 K
svchost.exe                   3340 Services                   0      5,576 K
svchost.exe                   3348 Services                   0      4,348 K
NVDisplay.Container.exe       3428 Services                   0     25,300 K
AsusCertService.exe           3528 Services                   0      2,956 K
svchost.exe                   3536 Services                   0     50,248 K
svchost.exe                   3600 Services                   0      3,188 K
svchost.exe                   3616 Services                   0     19,216 K
svchost.exe                   3624 Services                   0      5,228 K
Memory Compression            3752 Services                   0  12,38,832 K
svchost.exe                   3816 Services                   0      6,852 K
svchost.exe                   3868 Services                   0      5,432 K
svchost.exe                   3876 Services                   0      6,180 K
svchost.exe                   3972 Services                   0     14,336 K
svchost.exe                   4084 Services                   0      2,256 K
svchost.exe                   4812 Services                   0      5,724 K
svchost.exe                   3088 Services                   0      4,652 K
svchost.exe                   3744 Services                   0     23,128 K
svchost.exe                   4188 Services                   0     17,136 K
AsusOptimization.exe          41B4 Services                   0      5,404 K
svchost.exe                   41B6 Services                   0      8,648 K
spoolsv.exe                   43B4 Services                   0      8,888 K
svchost.exe                   44B4 Services                   0     13,368 K
svchost.exe                   45B6 Services                   0      4,980 K
svchost.exe                   46B4 Services                   0      5,588 K
ArmouryCrate.Service.exe      4936 Services                   0     36,744 K
ArmouryCrateControlInterf     4948 Services                   0      7,684 K
AsusSoftwareManager.exe       49B4 Services                   0     17,220 K
AsusSystemDiagnosis.exe       4988 Services                   0      4,284 K
AsusSwitch.exe                49B8 Services                   0      3,588 K
AsusSystemAnalysis.exe        5008 Services                   0      9,244 K
AsusAppService.exe            5028 Services                   0     10,312 K
C-MediaAudioService.exe       5048 Services                   0      2,676 K
AuraWallpaperService.exe      5060 Services                   0     13,376 K
svchost.exe                   5092 Services                   0     16,488 K
DAX3API.exe                   5188 Services                   0     11,508 K
svchost.exe                   3864 Services                   0     40,332 K
GameSDK.exe                   41B2 Services                   0      3,356 K
svchost.exe                   3660 Services                   0     46,124 K
ipf_uf.exe                    44B6 Services                   0      3,576 K
WMIRegistrationService.ex     4500 Services                   0      6,068 K
RstMwService.exe              5128 Services                   0      4,344 K
nvcontainer.exe               5136 Services                   0     21,536 K
LightingService.exe           5144 Services                   0     18,540 K
MpDefenderCoreService.exe     5152 Services                   0     17,960 K
OfficeClickToRun.exe          44A8 Services                   0     31,928 K
svchost.exe                   5168 Services                   0      7,480 K
servicehost.exe               5196 Services                   0     27,484 K
svchost.exe                   5208 Services                   0     16,848 K
MsMpEng.exe                   5216 Services                   0  2,55,804 K
svchost.exe                   5232 Services                   0      2,744 K
svchost.exe                   5244 Services                   0      5,852 K
ROGLiveService.exe            5260 Services                   0     11,184 K
svchost.exe                   5268 Services                   0     16,432 K
svchost.exe                   5276 Services                   0      3,652 K
ipfsvc.exe                    5284 Services                   0      3,828 K
svchost.exe                   5356 Services                   0      6,252 K
jhi_service.exe               6000 Services                   0      2,912 K
NVDisplay.Container.exe       6836 Console                    1     28,472 K
WmiPrvSE.exe                  7480 Services                   0     11,488 K
WmiPrvSE.exe                  7420 Services                   0     25,180 K
uihost.exe                    8568 Console                    1      9,364 K
```

EVIDENCE PRESERVATION

Methods to maintain evidence integrity during forensic investigation.
Ensuring data remains unaltered and admissible for analysis and legal proceedings.

Shutdown Prevention

Keep system powered to preserve volatile data from loss.

Workstation Recording

Photograph screen and environment for chain of custody documentation.

Disk Integrity

Verify disk status and health before creating forensic images.

Safe Handling

Use write blockers and careful procedures to avoid altering evidence.

ACQUISITION FORENSIC IMAGING

Creating verified forensic images to preserve evidence integrity.

Image Creation

Hash Verification

Value Matching



DATA EXAMINATION

Explore acquired forensic images to identify and extract evidential artefacts that reveal suspicious file access patterns and user activity traces



File Listing

Catalog all files accessed or modified suspiciously.



Deleted Pointers

Recover deleted file metadata and directory entries from disk.



History Review

Analyze user activity logs and file access histories for patterns.



Browser Artefacts

Extract internet history and cached data for investigation clues.

FORENSIC ANALYSIS
RECONSTRUCT THE TIMELINE

Interpreting findings to build a chronological sequence of the suspicious file access incident.

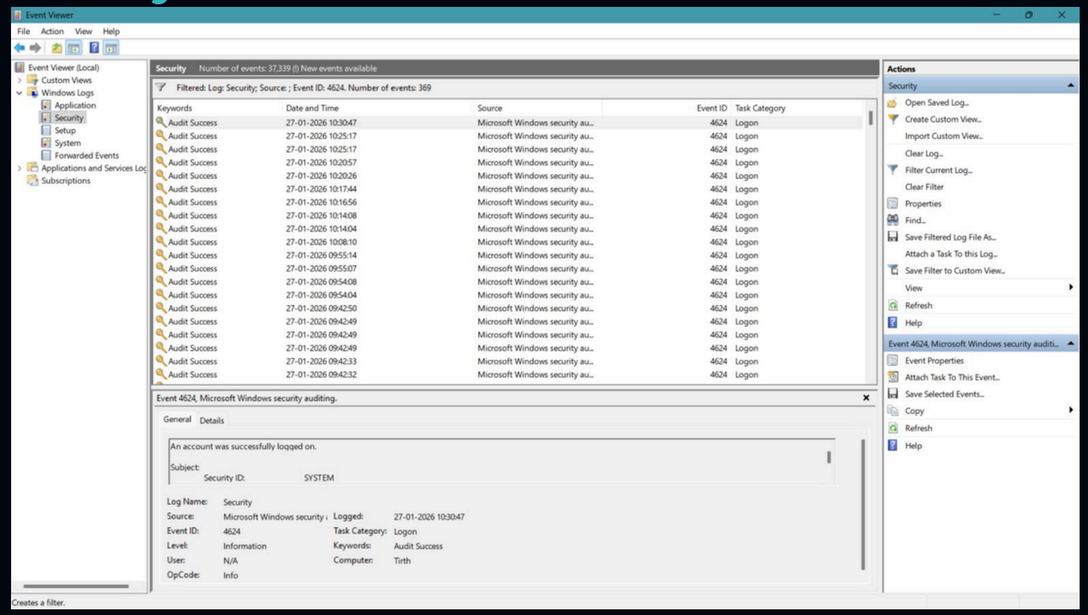




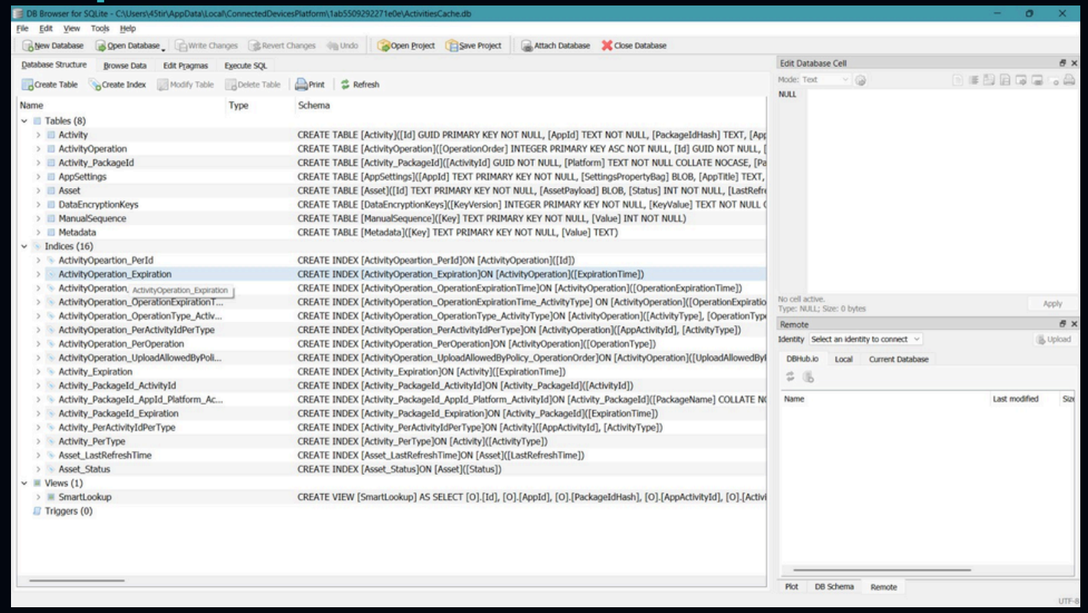
Log Analysis

- System event logs analyzed
- Login and logout activity checked
- Security events identified
- Suspicious login patterns detected
- User activity verified

History

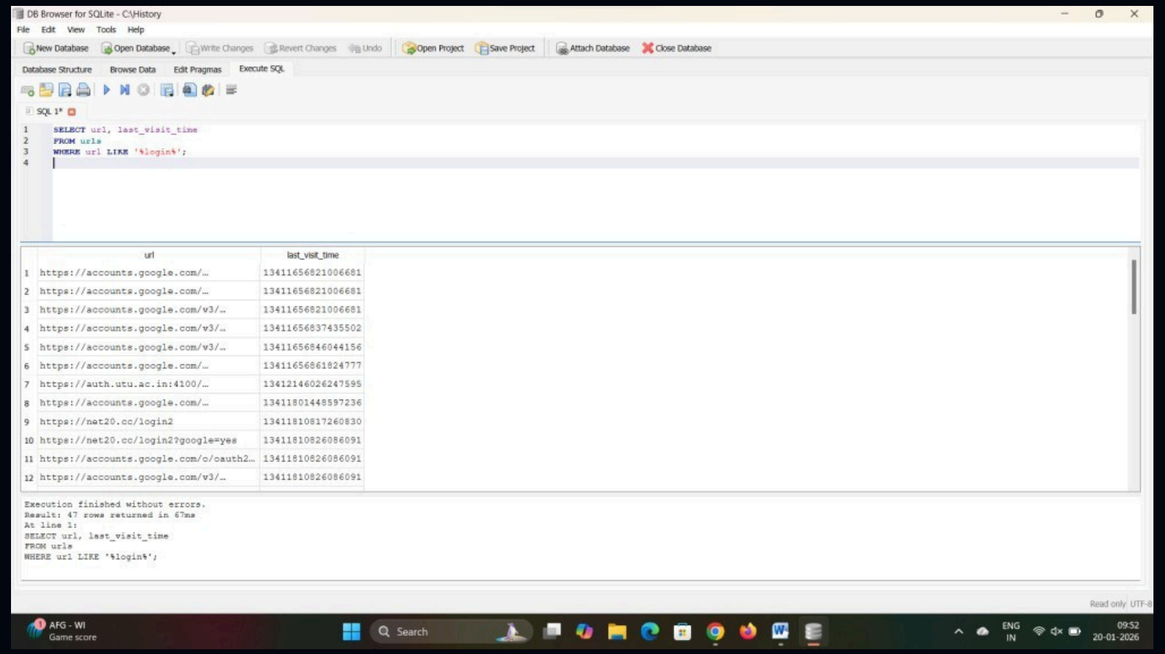


Suspicious Activities

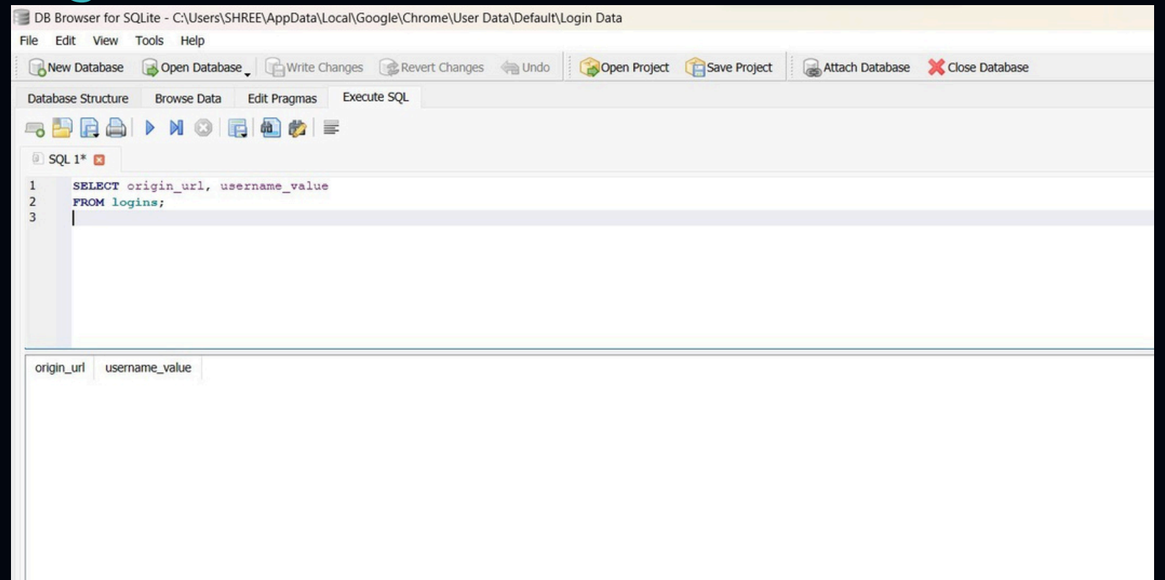


Authentication Review

- User accounts analyzed
- Login sessions examined
- Access credentials reviewed
- Unauthorized access identified
- User identity verified

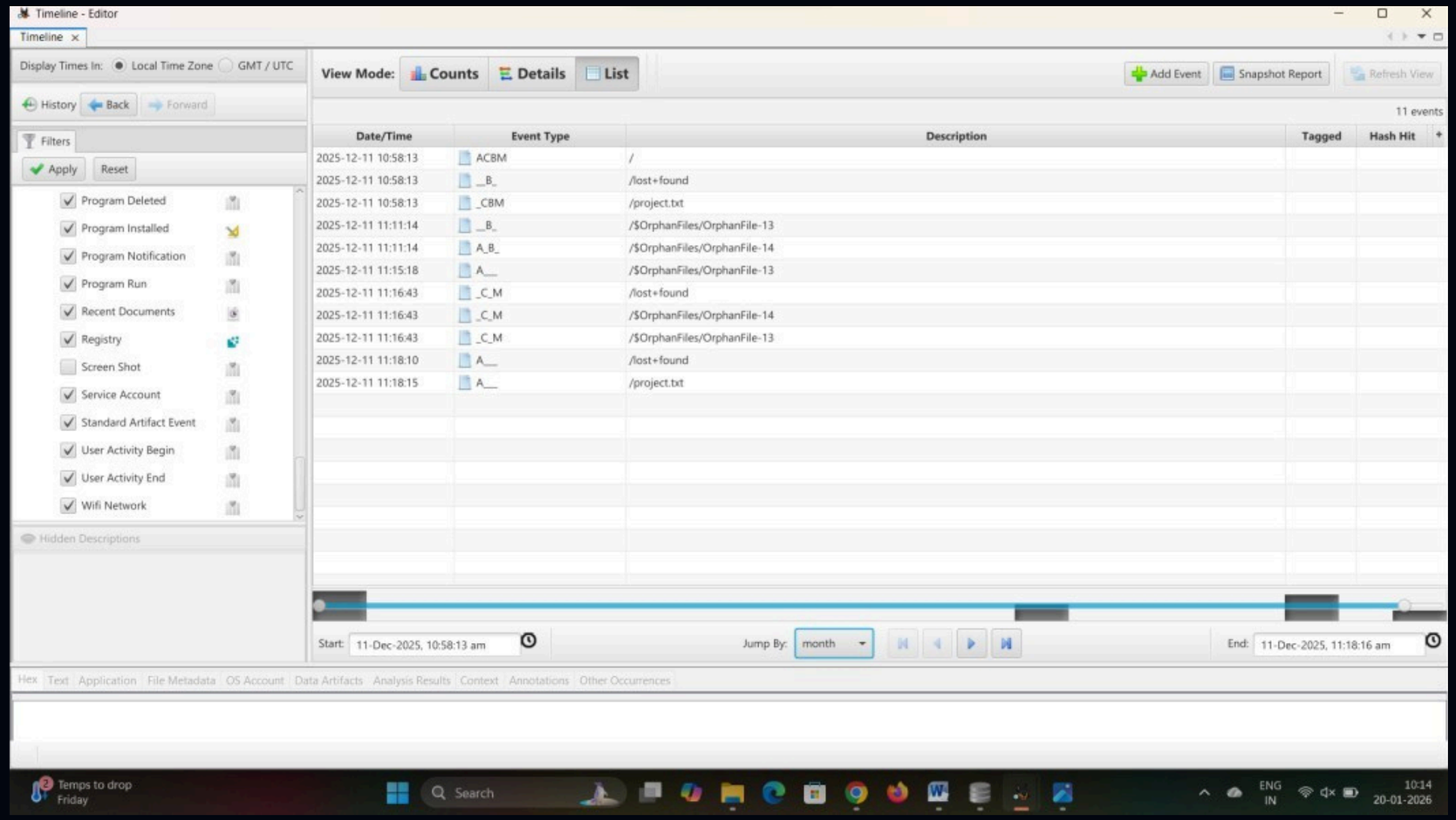


Login Data



Timeline Build

- Sequence of events created
- File access time analyzed
- Login and activity correlated
- Suspicious actions ordered chronologically
- Complete incident flow identified



FORENSIC REPORTING

Comprehensive documentation of all investigation steps, methodologies employed, and findings discovered throughout the forensic examination.

Evidence Screenshots

Visual documentation capturing key system states, file access patterns, and suspicious activity indicators for clear presentation of findings.



Activity Report

Detailed summary of investigation process and results.



Hash Proof

MD5/SHA1 hash values proving evidence integrity maintained.



Chain of Custody

Complete documentation trail for legal admissibility.

ARCHIVING EVIDENCE

Secure long-term storage of forensic evidence and case records ensures data integrity and availability for future legal proceedings.

Evidence Storage

Store forensic images and extracted data on secure, encrypted media with physical or digital access controls in a secure locker environment.



Database Entry

Log case metadata in forensic records management system.



Long-term Retention

Plan for future access and legal requirements compliance.



Chain of Custody

Maintain documentation of all evidence handling activities.

CASE CONCLUSION

Suspicious file access was confirmed through timeline reconstruction and preserved digital evidence. No external breach was detected—activity traced to local user actions.

Key Takeaways

The forensic process maintained evidence integrity throughout, enabling clear and defensible reporting. This case underscores the importance of prompt, methodical digital investigation.



THANK YOU FOR ATTENDING OUR CASE STUDY



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